

Short-Term Weather Prediction Using Deep Learning

Aqsa Mohsin

Course: Practical Deep Learning for Climate Data

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1 Introduction

Accurate weather forecasting plays a crucial role in agriculture, public safety, transportation, and energy management. Traditional numerical weather prediction methods are computationally expensive. This project explores deep learning alternatives—specifically LSTM and GRU—for short-term temperature forecasting using time-series data.

2 Dataset

We use the Daily Minimum Temperatures dataset for Melbourne, Australia, spanning 1981 to 1990. Preprocessing involved MinMax normalization and conversion into sequences using a 7-day sliding window. **Split:** 80% of the data used for training, 20% for testing (chronological order).

3 Methodology

The experiment involved two RNN-based deep learning models: LSTM and GRU. Both were trained to predict the next day's temperature using the past 7 days. Two baseline models were used for comparison: a Persistence model, which uses today's value as tomorrow's prediction, and a Climatology model that predicts based on historical averages.

These models were compiled using the Adam optimizer and trained with a mean squared error (MSE) loss function. The training process was conducted for 20 epochs with a batch size of 32. These choices offered a balance between learning speed and stability during convergence.

Why These Models and Metrics?

LSTM and GRU are well-suited for time-series because they can capture temporal dependencies using memory cells. MAE and RMSE are used to evaluate model accuracy. MAE provides an average error in degrees, while RMSE penalizes larger errors, making it sensitive to significant deviations in forecasts.

4 Results and Discussion

The models were evaluated using MAE and RMSE. Below are the updated results:

Model	MAE	RMSE
LSTM	1.8491	2.3343
GRU	1.7849	2.2634
Persistence	1.9551	2.4826
Climatology (Seasonal)	2.1107	2.7046

Table 1: Updated performance comparison of all models

Discussion: The GRU model achieved the best performance, slightly outperforming LSTM in both MAE and RMSE. Both deep learning models significantly outperformed the baseline models, demonstrating their effectiveness in capturing short-term temporal trends. The Persistence model, while simple, performed reasonably well. The enhanced Climatology model, using day-of-year averages, provided a seasonal baseline but could not adapt to short-term fluctuations.

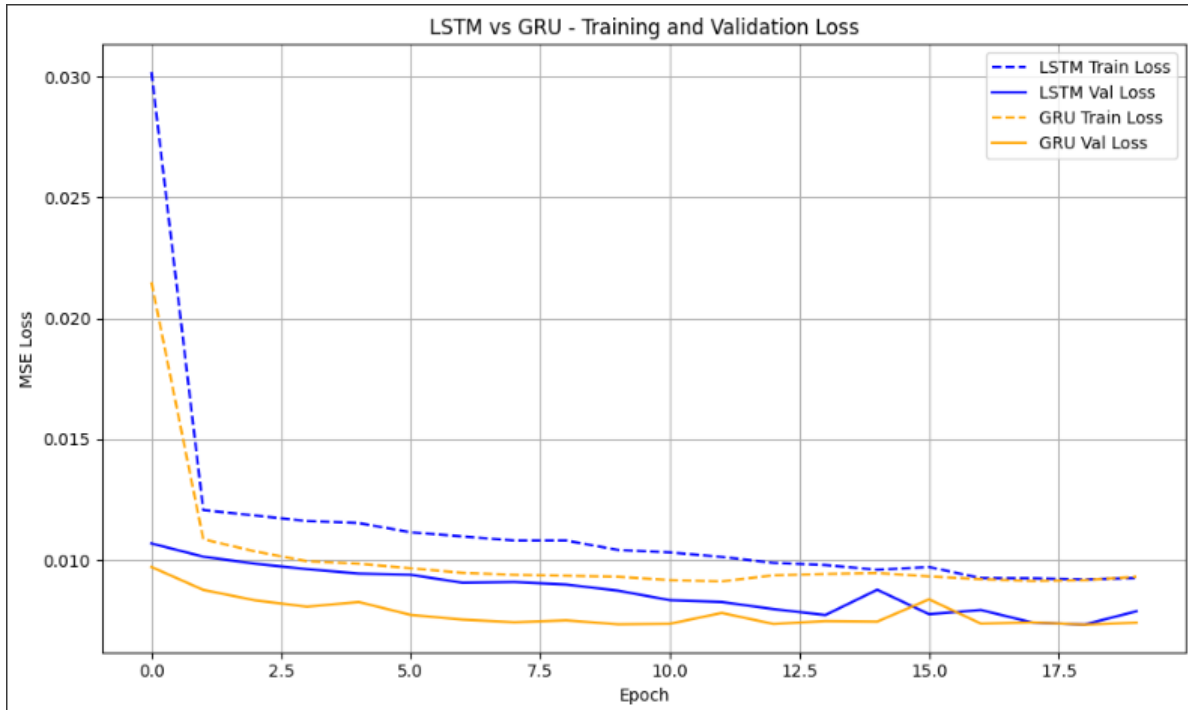


Figure 1: Training and Validation Loss Comparison: LSTM vs GRU. Dashed lines represent training loss, solid lines represent validation loss.

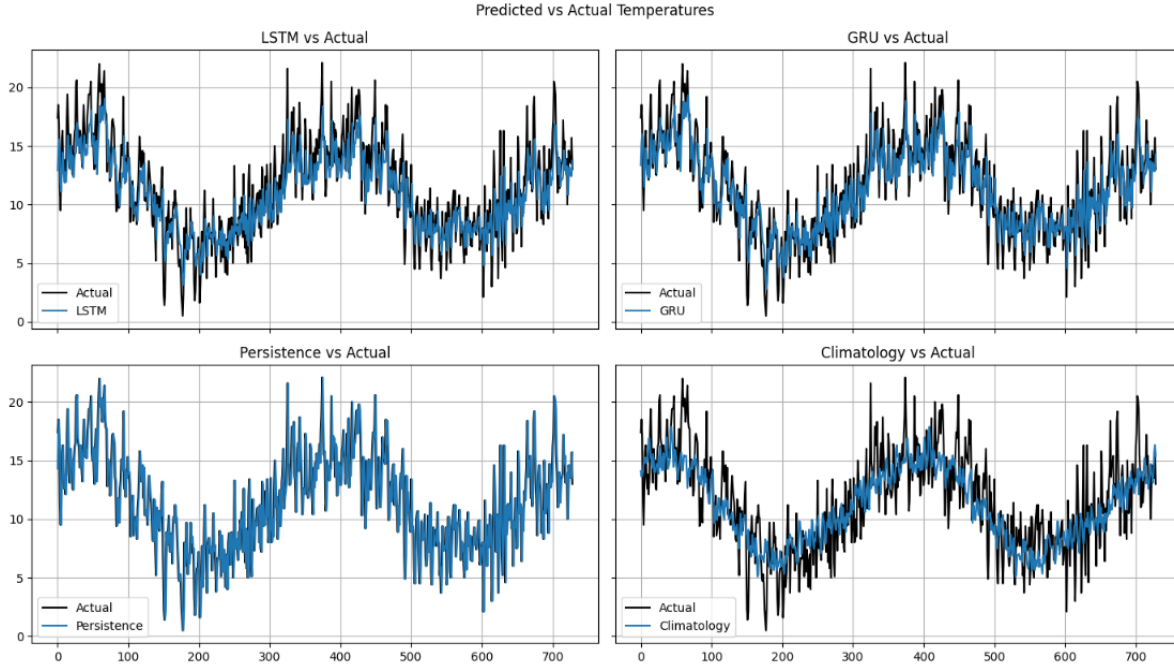


Figure 2: Predicted vs Actual Temperatures for All Models — Including Enhanced Climatology with Seasonal Trend

5 Conclusion

This project demonstrated that RNN-based models are effective for short-term temperature forecasting. LSTM and GRU outperformed simple baselines like persistence and climatology, with GRU showing slightly better convergence and generalization. The enhanced climatology approach brought seasonal understanding to baseline prediction.

Future work could include:

- Adding multivariate inputs (humidity, wind speed, etc.)
- Exploring longer historical input windows
- Investigating transformer-based models for sequence modeling

6 References

- Brownlee, J. Daily Min Temperature Dataset. <https://raw.githubusercontent.com/jbrownlee/Datasets/master/daily-min-temperatures.csv>
- TensorFlow. <https://www.tensorflow.org/>
- Scikit-learn. <https://scikit-learn.org/>